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SE2020 – Web Technologies and Mobile

Lecture 04

Html DOM

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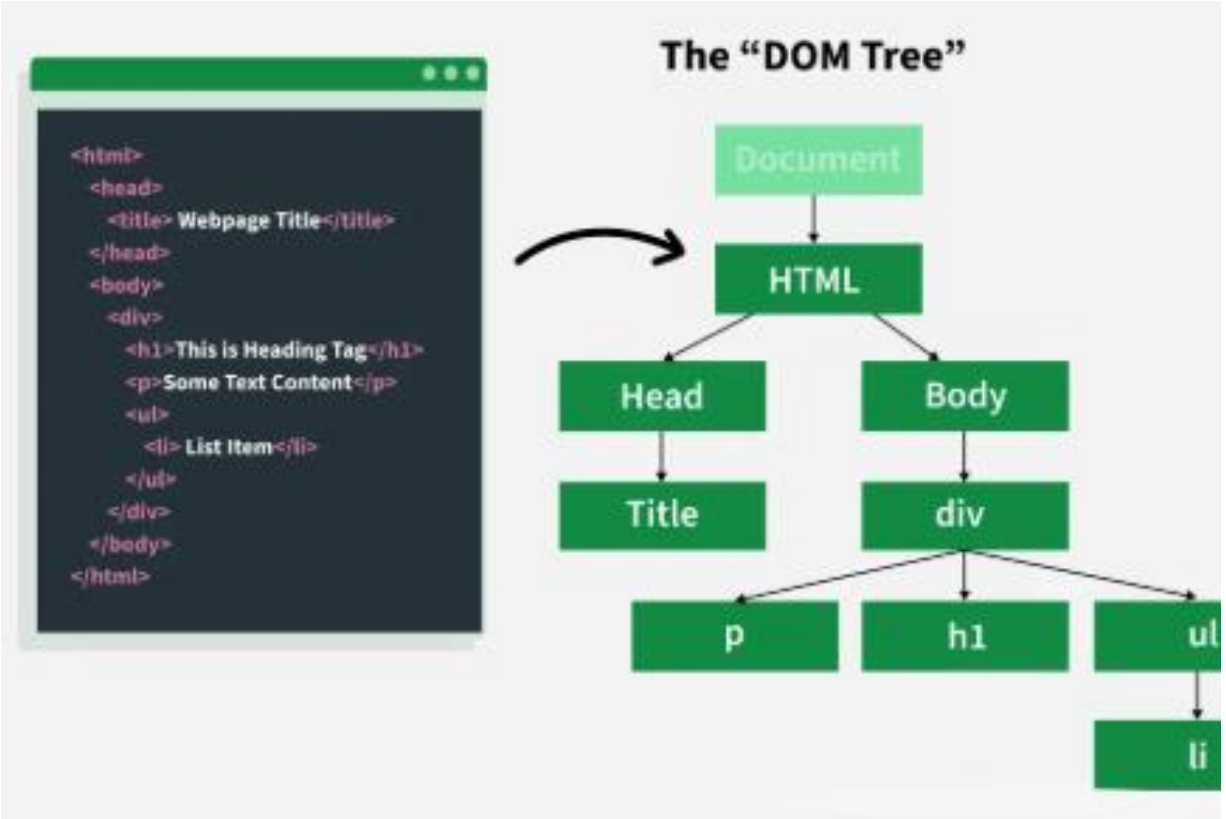
FACULTY OF COMPUTING

Introduction to the HTML DOM (Document Object Model)

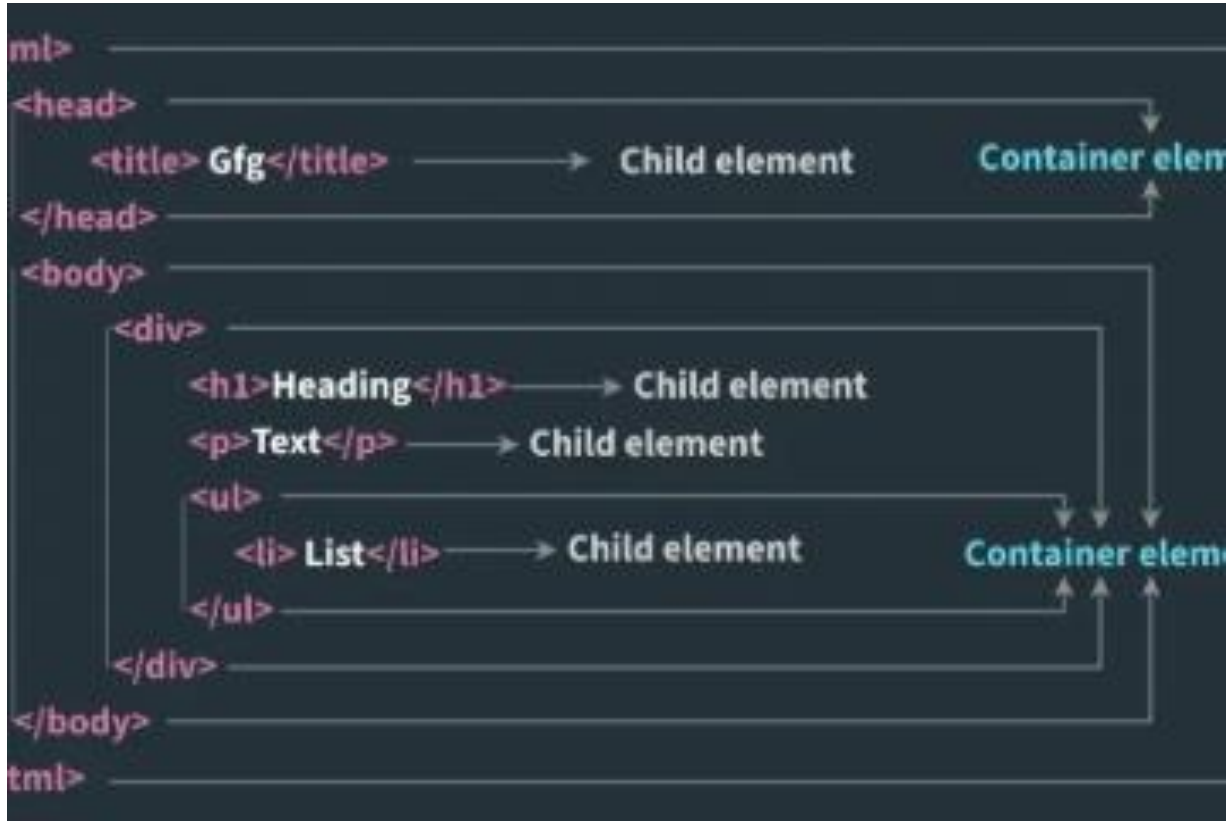
- We will only cover aspects of the DOM that we need to read/change form element values/settings, document content and document style attributes
- The Document Object Model is a platform- and language-neutral interface that will allow programs and scripts to dynamically access and update the content, structure and style of documents.
- The document can be further processed and the results of that processing can be incorporated back into the presented page. This is an overview of DOM-related materials here at W3C and around the web. (W3C)
- The HTML DOM model is constructed as **a tree of Objects**, created by the browser, so we can quickly find HTML elements using JavaScript

DOM


The "DOM Tree"
<html>
<head>
<title> Webpage Title</title>
</head>
<body>

<h1>This is Heading Tag</h1>
Some Text Content
<ul>
<li> List Item</li>
</ul>

</body>
</html>
Document
HTML
Head
Body
Title
div
p
h1
ul
li


ml>
<head>
<title> Gfg</title>
Child element
Container elem
</head>
<body>

<h1>Heading</h1>
Child element
Text
Child element
<ul>
<li> List</li>
Child element
Container elem
</ul>

</body>
tml>

JavaScript and the DOM

JavaScript can be used to

- ❖change HTML elements in the page
- ❖change HTML attributes in the page
- ❖change the CSS styles in the page
- ❖react to the events in the page

in order to do that JavaScript should find element in an html page.

Finding HTML Elements

- ❖finding HTML elements by id
- ❖ finding HTML elements by tag name
- ❖ finding HTML elements by class name

DOM – Methods – in camel case

Document Object Methods

Method	Description	W3C
close()	Closes the output stream previously opened with document.open()	Yes
getElementsByName()	Accesses all elements with a specified name	Yes
getElementById()	Accesses the element with the specified id	Yes
getElementsByClassName()	Accesses all elements with a specified class name	Yes
getElementsByTagName()	Accesses all elements with a specified tag name	Yes
open()	Opens an output stream to collect the output from document.write() or document.writeln()	Yes
write()	Writes HTML expressions or JavaScript code to a document	Yes
writeln()	Same as write(), but adds a newline character after each statement	Yes

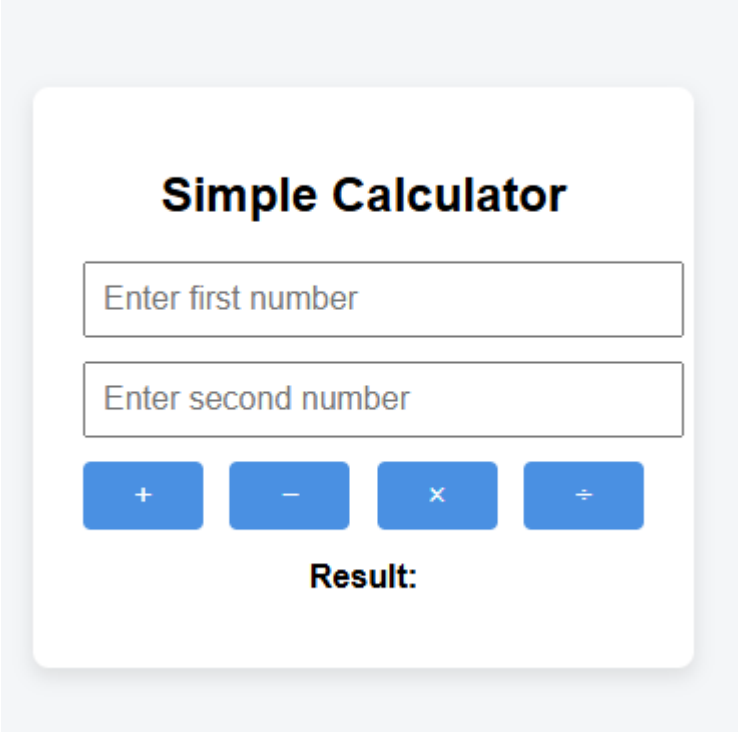
getElementById()

```
<label>Enter first number</label>
<input type="text" id="num1"/>
<br>
<label>Enter second number</label>
<input type="text" id="num2"/>
<button onclick="addNumber()">+ </button>
<p id="sumValue"></p>
<script>
  function addNumber(){

    let num1=document.getElementById("num1").value;
    let num2=document.getElementById("num2").value;
    let results=parseInt(num1)+ parseInt(num2);

   document.getElementById("sumValue").innerHTML =num1 + " + " + num2 + " = "+results;
  }
```

Now shall we create a simple calculator to do addition Multifaction , division and subtraction



getElementByTagName()

```
<body>
  <!-- Change color html code - getElementByTagName-->
  <p>This is the First paragraph</p>
  <p>This is the Second paragraph</p>
  <p>This is the Third paragraph</p>
  <button onclick="changeColor()">Change Text Color</button>
  <script>
    function changeColor() {
      let paragraphs = document.getElementsByTagName("p");

      paragraphs[0].style.color = "pink";
      paragraphs[1].style.color = "aqua";
      paragraphs[2].style.color = "purple";
    }
  </script>
```

This is the First paragraph

This is the Second paragraph

This is the Third paragraph

Change Text Color

Now Shall we change html content Dynamically as follows with a button click

Updated first paragraph

Updated second paragraph

Updated third paragraph

Change Text Color | Change Text Content

getElementByName()

- Returns a **NodeList**
- Mostly used for **form inputs**
- Access using **index**

```html
<!-- getElementByName -->
  <form>
  <input type="text" name="username" placeholder="Enter username">
  <input type="text" name="username" placeholder="Re-enter username">
  <button type="button" onclick="showName()">Show</button>
</form>
<p id="output"></p>
<script>
function showName() {
  let inputs = document.getElementsByName("username");
  document.getElementById("output").textContent = inputs[0].value;
}

</script>
```

```html
<h3>Select Gender</h3>

<input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female

<br><br>
<button onclick="showGender()">Show Gender</button>

<p id="result"></p>
```

```js
function showGender() {
  let gender = document.getElementsByName("gender");

  for (let i = 0; i < gender.length; i++) {
    if (gender[i].checked) {
      document.getElementById("result").textContent =
        "Selected: " + gender[i].value;
    }
  }
}
```

Try this code

getElementByClassName()

- Returns an **HTMLCollection (**classlist**)**
- Very fast
- Live collection (updates automatically)

```html
<!-- getElementByClassName-->
 <h3>Choose a Course</h3>

<div class="course">Web Development</div>
<div class="course">Mobile App Development</div>
<div class="course">Data Science</div>

<button onclick="highlight()">Highlight Courses</button>

<script>
function highlight() {
  let courses = document.getElementsByClassName("course");
  for (let i = 0; i < courses.length; i++) {
    courses[i].classList.add("active");
  }
}
</script>
```

```style
<style>
.course {
  padding: 10px;
  margin: 5px 0;
  border: 1px solid #ccc;
}

.active {
  background-color: #d1e7ff;
  font-weight: bold;
}
</style>
```

Choose a Course

Web Development

Mobile App Development

Data Science

Highlight Courses

Choose a Course

Web Development

Mobile App Development

Data Science

Highlight Courses

Try this code

Changing HTML Style with the HTML DOM Style Object

Text properties

Property	Description	W3C
color	Sets or returns the color of the text	Yes
direction	Sets or returns the text direction	Yes
font	Sets or returns font-style, font-variant, font-weight, font-size, line-height, and font-family in one declaration	Yes
fontFamily	Sets or returns the font face for text	Yes
fontSize	Sets or returns the font size of the text	Yes
fontSizeAdjust	Sets or returns the font aspect value	Yes
fontStyle	Sets or returns whether the style of the font is normal, italic or oblique	Yes
fontVariant	Sets or returns whether the font should be displayed in small capital letters	Yes
fontWeight	Sets or returns the boldness of the font	Yes
letterSpacing	Sets or returns the space between characters in a text	Yes
lineHeight	Sets or returns the distance between lines in a text	Yes

Background properties

Property	Description	W3C
background	Sets or returns all the background properties in one declaration	Yes
backgroundAttachment	Sets or returns whether a background-image is fixed or scrolls with the page	Yes
backgroundColor	Sets or returns the background-color of an element	Yes
backgroundImage	Sets or returns the background-image for an element	Yes
backgroundPosition	Sets or returns the starting position of a background-image	Yes
backgroundRepeat	Sets or returns how to repeat (tile) a background-image	Yes

HTML DOM Events – (all in simple letters)

- HTML DOM events allow JavaScript to register different event handlers on elements in an HTML document.
- Events are normally used in combination with functions, and the function will not be executed before the event occurs (such as when a user clicks a button).
- Some common uses of events
  - When a user clicks the mouse
  - When a web page has loaded
  - When an image has been loaded
  - When the mouse moves over an element
  - When an input field is changed
  - When an HTML form is submitted
  - When a user strokes a key

Key Board Events

Attribute	Description
onkeydown	The event occurs when the user is pressing a key
onkeypress	The event occurs when the user presses a key
onkeyup	The event occurs when the user releases a key

Form Events

Attribute	Description
onblur	The event occurs when a form element loses focus
onchange	The event occurs when the content of a form element, the selection, or the checked state have changed (for <input>, <select>, and <textarea>)
onfocus	The event occurs when an element gets focus (for <label>, <input>, <select>, textarea>, and <button>)
onreset	The event occurs when a form is reset
onselect	The event occurs when a user selects some text (for <input> and <textarea>)
onsubmit	The event occurs when a form is submitted

onkeydown

```html
<!--keyboard onkeydown -->
<h3>Press any key</h3>
<input type="text" onkeydown="showKey(event)">
<p id="result"></p>
<script>
  function showKey(e) {
    document.getElementById("result").textContent =
      "Key pressed: " + e.key;
  }
</script>
```

onchange

```html
<h3>Select your city</h3>
<select onchange="showCity(this.value)">
  <option value="">-- Select City --</option>
  <option value="Colombo">Colombo</option>
  <option value="Kandy">Kandy</option>
  <option value="Galle">Galle</option>
</select>
<p id="result"></p>
<script>
  function showCity(city) {
    document.getElementById("result").textContent =
      "You selected: " + city;
  }
</script>
```

Image Replacement

- Image Properties

Property	Description
align	Sets or returns the value of the align attribute of an image
alt	Sets or returns the value of the alt attribute of an image
border	Sets or returns the value of the border attribute of an image
complete	Returns whether or not the browser is finished loading an image
height	Sets or returns the value of the height attribute of an image
hspace	Sets or returns the value of the hspace attribute of an image
longDesc	Sets or returns the value of the longdesc attribute of an image
lowsrc	Sets or returns a URL to a low-resolution version of an image
name	Sets or returns the name of an image
src	Sets or returns the value of the src attribute of an image
useMap	Sets or returns the value of the usemap attribute of an image
vspace	Sets or returns the value of the vspace attribute of an image
width	Sets or returns the value of the width attribute of an image

A Simple Example

- Changing the URL of the image using the src property causes the new image to be loaded gradually onto the page.
- Replacement is not immediate.

```
<img src="simple0.jpg" width="257" height="75" name="sample" alt="simple image">

<form>
<input type="button" value="simple0.jpg" onclick="document.images[0].src = 'simple0.jpg'">
<input type="button" value="simple1.jpg" onclick="document.images[0].src = 'simple1.jpg'">
</form>
```

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Thank You